

Adiabatic invariant

Y13 (2013) — English translation

A block of mass M moves along a smooth horizontal plane with a speed v_0 towards the wall. A small ball of mass m is at rest at a distance L from the wall. A collision occurs.

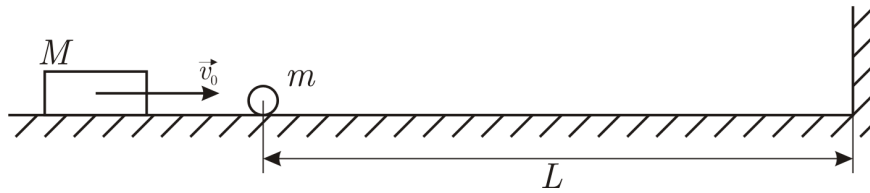


Figure 1: Figure 1.

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| A1 | Determine the minimum distance at which the block will approach the wall, and how many collisions between the ball and the block will occur. | 0 |
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All impacts can be considered absolutely elastic and $M \gg m$.

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| A2 | Find the adiabatic invariant that describes the motion of the ball (after the first collision with the block). | 0 |
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Source: <https://pho.rs/p/3997>